

Yissum Research Intelligence Report — Exact & Social Sciences — July 2026

2 paper(s) · Exact & Social Sciences · Monthly · July 2026 · Generated July 22, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This month's digest showcases compelling research at the intersection of fundamental science and applied technology. One featured project explores advanced quantum dynamics to enhance the efficiency of next-generation optoelectronic devices, promising breakthroughs in clean energy and high-performance electronics. Simultaneously, another initiative unveils a novel AI-driven approach to combat disinformation and improve public trust in scientific information, with significant implications for digital health and effective communication strategies. These innovations highlight Hebrew University's commitment to both deep tech advancement and practical, societal-impact solutions.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- Eran Rabani — Materials, Clean Energy, Quantum
- Anat Lande-Brenner — Software/AI, Other

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Full Quantum and Mixed Quantum-Classical Dynamics of Hot Exciton Cooling **21/50** in Semiconductor Nanocrystals.

"Quantum simulations reveal key insights for more efficient next-gen solar cells and quantum dot LEDs."

Main researcher: Eran Rabani

Department of Chemistry, University of California, Berkeley, California 94720, United States.; Fritz Haber Center for Molecular Dynamics, The Institute of Chemistry and The Institute of Applied Physics, The Hebrew University of Jerusalem, Jerusalem 91904, Israel.; Materials Sciences Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States.

Published in The journal of physical chemistry letters · 2026 Jul 14

WHY NOW

The global demand for sustainable energy and advanced displays drives intense R&D for more efficient optoelectronic materials. Optimizing hot carrier dynamics presents a critical, near-term opportunity to unlock performance improvements in existing and emerging device architectures.

COMMERCIAL ANGLE

Improving the efficiency of next-generation optoelectronic devices, such as high-efficiency solar cells and quantum dot LEDs, by optimizing hot carrier extraction.

COMMERCIALISATION METRICS

Novelty

6/10

The work applies established quantum and mixed quantum-classical dynamics frameworks to the specific problem of hot exciton cooling, providing incremental refinement rather than a groundbreaking new physical paradigm.

Commercial Potential

3/10

The paper focuses on fundamental theoretical dynamics of exciton cooling, which is a basic physics exploration rather than a direct application or device architecture. While it informs the design of future optoelectronics, there is no clear path to a specific product or licensable technology presented.

Market Size

7/10

The work addresses fundamental carrier dynamics in semiconductor nanocrystals, which are critical for multi-billion dollar markets in next-generation LEDs, quantum dot displays, and high-efficiency photovoltaics.

Tech Readiness

2/10

The paper focuses on theoretical modeling and computational dynamics of quantum processes, representing basic fundamental research (TRL 2) rather than an applied prototype or commercial product.

IP Strength

3/10

The paper describes fundamental theoretical modeling of quantum dynamics, which is generally considered non-patentable scientific discovery rather than a protectable industrial invention.

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2. When scientific sources fail to convince: evidence-oriented thinking enhances **17/50** the effectiveness of science-based corrections.

"AI-driven cognitive nudges prove more effective than citations in combating disinformation and building trust."

Main researcher: Anat Lande-Brenner · anat.lande@mail.huji.ac.il.

Department of Psychology, The Hebrew University of Jerusalem, Jerusalem, Israel.; Federmann Center for the Study of Rationality, The Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Scientific reports · 2026-06-25

WHY NOW

With the rapid spread of disinformation and declining public trust, there's an urgent market need for effective communication tools. This research offers a timely, empirically-validated strategy ready for integration into AI-powered digital health and public information platforms.

COMMERCIAL ANGLE

There is a commercial opportunity for developing AI-driven communication tools and digital health platforms that integrate 'cognitive nudges' to improve patient adherence and public trust in medical science.

COMMERCIALISATION METRICS

Novelty

5/10

While the study identifies an interesting psychological mechanism to mitigate the 'backfire effect,' the concept of using cognitive nudges to improve communication effectiveness is well-established in behavioral science. The novelty lies in the specific application to scientific disinformation rather than a fundamental breakthrough in cognitive theory or a new technical platform.

Commercial Potential

4/10

While the findings suggest a practical psychological intervention for information integrity, the technology is a soft 'cognitive nudge' rather than a hard, scalable deep-tech product or proprietary platform. Commercialization would likely rely on integration into existing social media moderation APIs or educational software rather than a standalone high-moat enterprise solution.

Market Size

4/10

While the problem of disinformation is globally pervasive, the research describes a behavioral intervention rather than a scalable deep-tech product or platform, limiting its direct commercial addressable market to niche digital health or social media moderation sectors.

Tech Readiness

2/10

This is basic behavioral science research conducted via simulated experiments rather than a deployed real-world intervention. While it provides theoretical frameworks for communication, it lacks the large-scale field testing or product integration required for higher TRL stages.

IP Strength

2/10

The innovation consists of cognitive prompting strategies and psychological interventions that are difficult to patent and lack any proprietary technological or biochemical moat. The findings represent behavioral insights rather than defensible, scalable deep-tech or life-science IP.

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